Behind the Supply Curve

Chapter 6
Behind the Supply Curve: Inputs and Costs

THE FARMER'S MARGIN



American farming practices (at left) or European farming practices (at right)? How intensively an acre of land is worked—a decision at the margin—depends on the price of wheat a farmer faces.

What you will learn in this chapter:

- The importance of the firm's **production function**, the relationship between quantity of inputs and quantity of output
- Why production is often subject to **diminishing returns to inputs**
- What the various forms of a firm's costs are and how they generate the firm's **marginal** and **average cost curves**
- Why a firm's costs may differ in the **short run** versus the **long run**
- How the firm's technology of production can generate **economies of scale**

The Production Function

A **production function** is the relationship between the quantity of inputs a firm uses and the quantity of output it produces.

Inputs and Output

A **fixed input** is an input whose quantity is fixed and cannot be varied.

A **variable input** is an input whose quantity the firm can vary.

The Production Function

Inputs and Output

The **long run** is the time period in which all inputs can be varied.

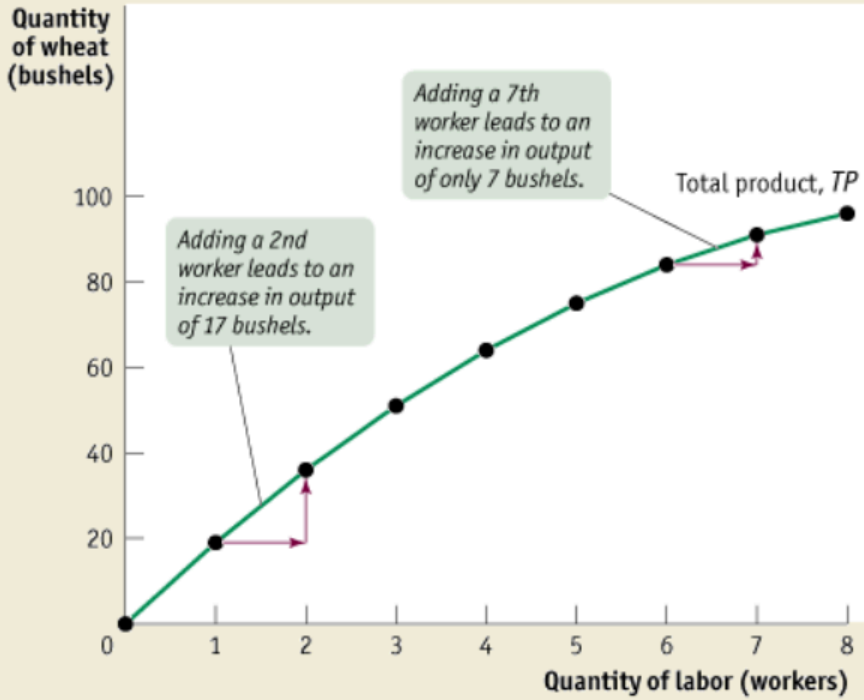
The **short run** is the time period in which at least one input is fixed.

The **total product curve** shows how the quantity of output depends on the quantity of the variable input, for a given quantity of the fixed input.

The Production Function

Inputs and Output

Figure 7-1 Production Function and Total Product Curve for George and Martha's Farm



Quantity of labor *L* (workers)	Quantity of wheat *Q* (bushels)	Marginal product of labor *MPL* = *Q*/*L* (bushels per worker)
0	0	
		19
1	19	
		17
2	36	
		15
3	51	
		13
4	64	
		11
5	75	
		9
6	84	
		7
7	91	
		5
8	96	

The table shows the production function, the relationship between the quantity of the variable input (labor, measured in number of workers) and the quantity of output (bushels of wheat) for a given quantity of the fixed input. It also calculates the marginal product of labor on George and Martha's farm. The total product curve shows the production function graphically. It slopes upward because more wheat is produced as more workers are employed. It also becomes flatter because the marginal product of labor declines as more and more workers are employed.

The Production Function

Inputs and Output

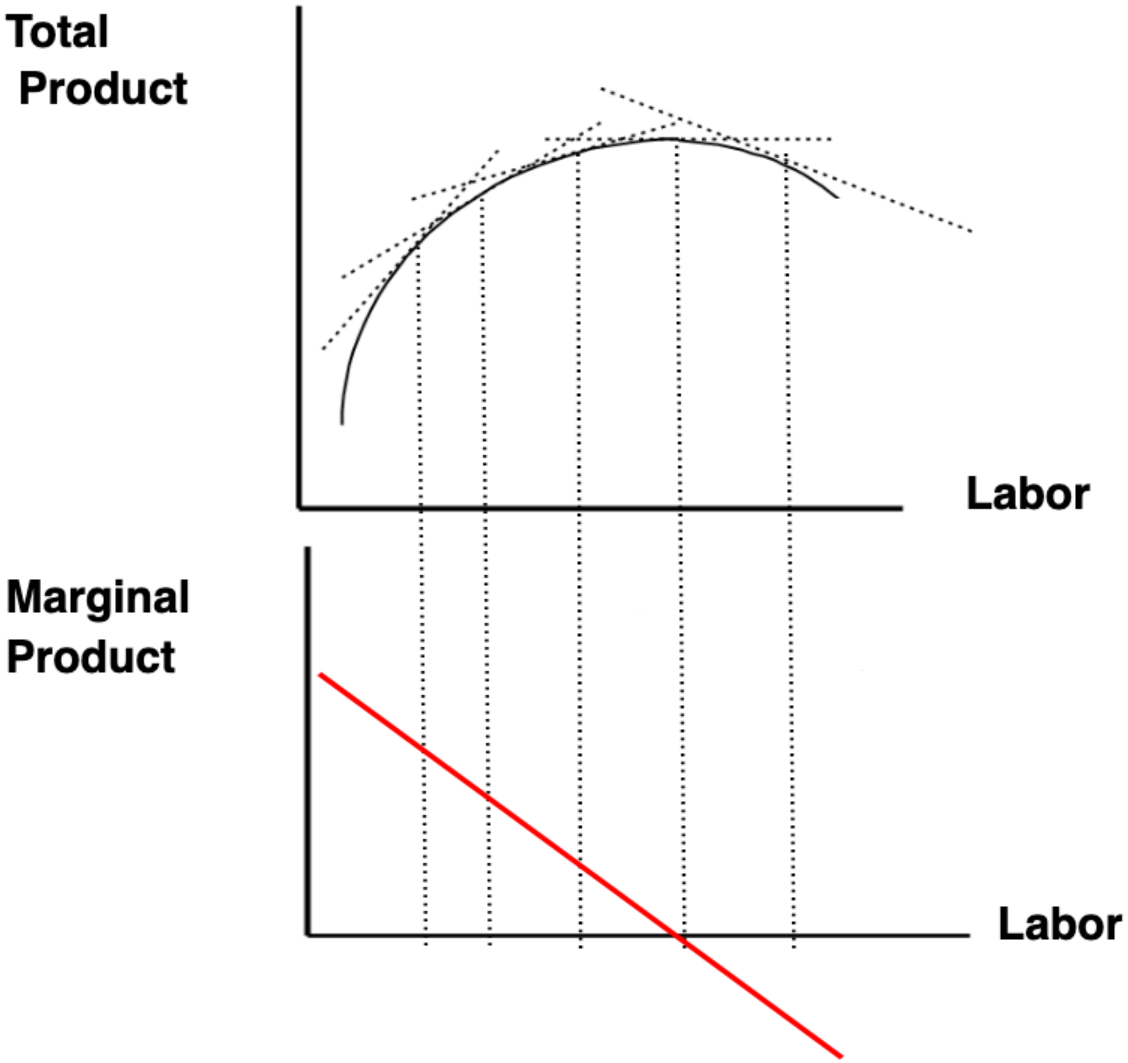
The **marginal product** of an input is the additional quantity of output that is produced by using one more unit of that input.

(7-1) Marginal product of labor $= \frac{\text{Change in quantity of output}}{\text{Change in quantity of labor}} =$ Change in quantity of output generated by one additional unit of labor

or

$$MPL = DQ/DL$$

Total Product and Marginal Product



While marginal product is positive, total product is increasing.

When marginal product is 0, total product does not change.

When marginal product is negative, total product decrease.

The Production Function

Inputs and Output

There are **diminishing returns to an input** when an increase in the quantity of that input, holding the levels of all other inputs fixed, leads to a decline in the marginal product of that input.

Diminishing returns = Marginal product is decreasing

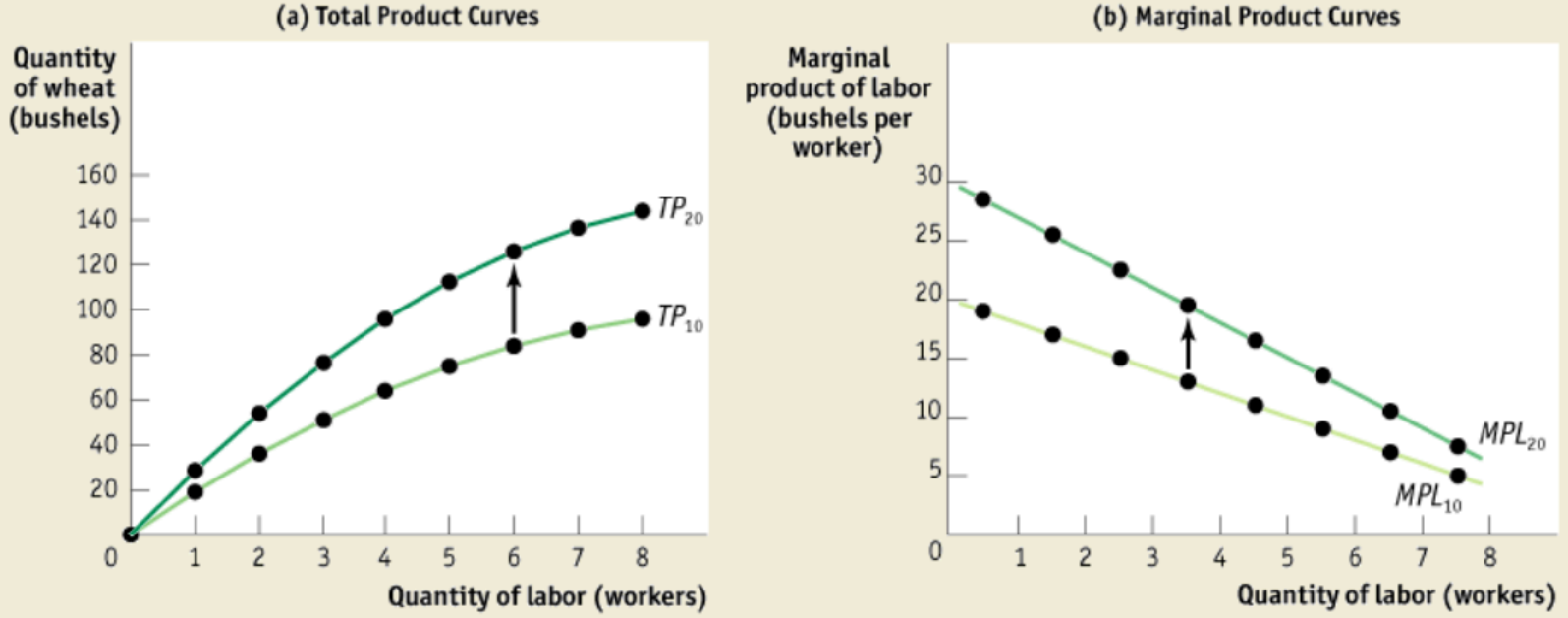
When f is the production function which depends on the input labor L,

Total production	$Q = f(L)$ is a increasing function on L
Marginal production	$Q' = f'(L) > 0$, but $Q'' = f''(L) < 0$

The Production Function

Inputs and Output

Figure 7-3 Total Product, Marginal Product, and the Fixed Input



This figure shows how the quantity of output—illustrated by the total product curve—and marginal product depend on the level of the fixed input. Panel (a) shows two total product curves for George and Martha's farm, TP_{10} when their farm is 10 acres and $TP20$ when it is 20 acres. With more land, each worker can produce more wheat. So an increase in the fixed input shifts the total product curve up from TP_{10} to TP_{20}. This also implies that the marginal product of each worker is higher when the farm is 20 acres than when it is 10 acres. As a result, an increase in acreage also shifts the marginal product of labor curve up from MPL_{10} to MPL_{20}. Panel (b) shows the marginal product of labor curves. Note that both marginal product of labor curves still slope downward due to diminishing returns.

Behind the Supply Curve: Inputs and Costs

An **explicit cost** is a cost that involves actually laying out money.

An **implicit cost** does not require an outlay of money; it is measured by the value, in dollar terms, of the benefits that are forgone.

Behind the Supply Curve: Inputs and Costs

From the Production Function to Cost Curves

A **fixed cost** is a cost that does not depend on the quantity of output produced. It is the cost of the fixed input.

A **variable cost** is a cost that depends on the quantity of output produced. It is the cost of the variable input.

The **total cost** of producing a given quantity of output is the sum of the fixed cost and the variable cost of producing that quantity of output.

(7-2) $\text{Total cost} = \text{Fixed cost} + \text{Variable cost}$

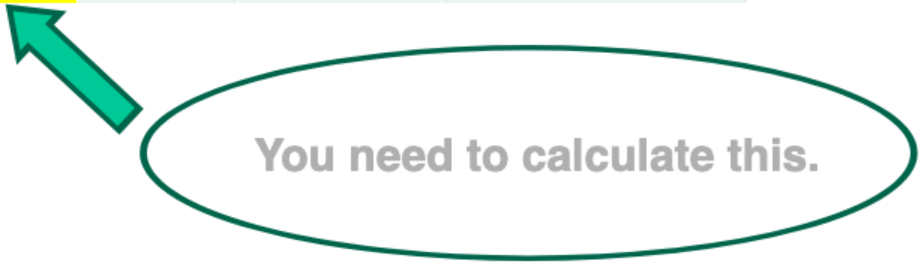
or

$$TC = FC + VC$$

The **total cost curve** shows how total cost depends on the quantity of output.

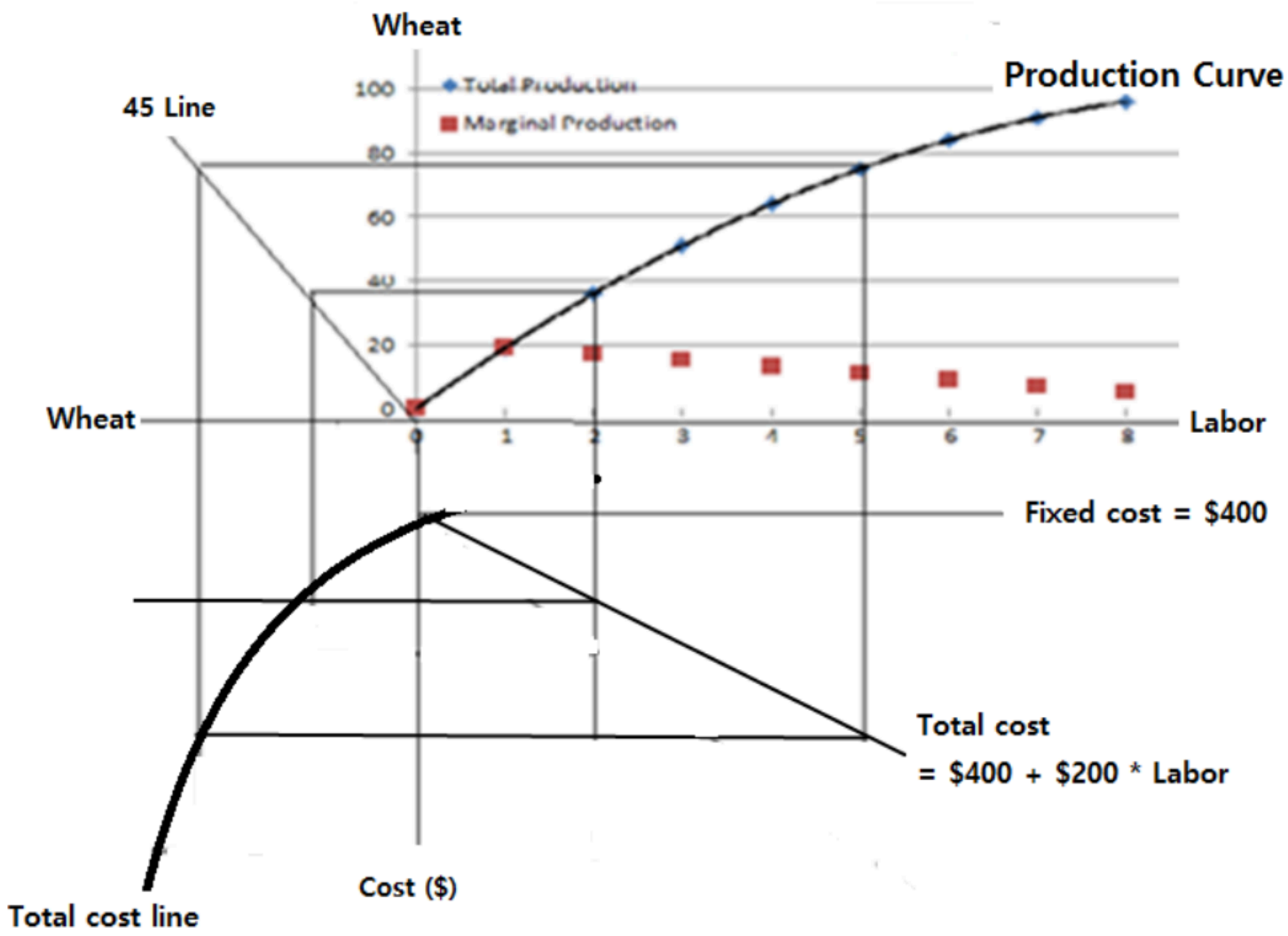
Total Cost

				Unit Labor Cost	Fixed Cost	
				$200	$400	
Point	Quantity of Labor	Quantity of wheat	Marginal production	Variable Cost	Fixed Cost	Total Cost
A	0	0	0	0	400	400
B	1	19	19	200	400	600
C	2	36	17	400	400	800
D	3	51	15	600	400	1000
E	4	64	13	800	400	1200
F	5	75	11	1000	400	1400
G	6	84	9	1200	400	1600
H	7	91	7	1400	400	1800
I	8	96	5	1600	400	2000



See the EXCEL file

From Production Curve to Total Cost Curve



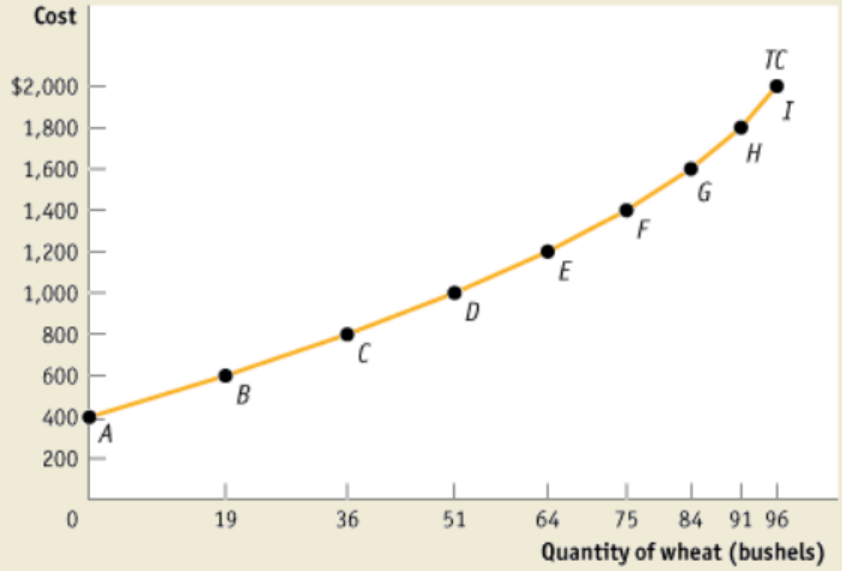
Behind the Supply Curve: Inputs and Costs

From the Production Function to Cost Curves

Figure 7-4

Total Cost Curve for George and Martha's Farm

The table shows the variable cost, fixed cost, and total cost for various output quantities on George and Martha's 10-acre farm. The total cost curve shows how total cost (measured on the vertical axis) depends on the quantity of output (measured on the horizontal axis). The labeled points on the curve correspond to the rows of the table. The total cost curve slopes upward because the number of workers employed, and hence total cost, increases as the quantity of output increases. The curve gets steeper as output increases due to diminishing returns to labor.



Point on graph	Quantity of labor *L* (workers)	Quantity of wheat *Q* (bushels)	Variable cost *VC*	Fixed cost *FC*	Total cost *TC* = *FC* + *VC*
A	0	0	$0	$400	$400
B	1	19	200	400	600
C	2	36	400	400	800
D	3	51	600	400	1,000
E	4	64	800	400	1,200
F	5	75	1,000	400	1,400
G	6	84	1,200	400	1,600
H	7	91	1,400	400	1,800
I	8	96	1,600	400	2,000

Two Key Concepts: Marginal Cost and Average Cost

Marginal Cost

The **marginal cost** of an activity is the additional cost incurred by doing one more unit of that activity.

(7-3) $$\text{Marginal cost} = \frac{\text{Change in total cost}}{\text{Change in quantity of output}} =$$ Change in total cost generated by one additional unit of output

or

$$MC = DTC / DQ$$

Two Key Concepts: Marginal Cost and Average Cost

Marginal Cost

TABLE 7-1

Costs at Ben's Boots

Quantity of boots *Q* (pairs)	Fixed cost *FC*	Variable cost *VC*	Total cost *TC = FC + VC*	Marginal cost of pair *MC = ΔTC/ΔQ*
0	$108	$0	$108	
				$12
1	108	12	120	
				36
2	108	48	156	
				60
3	108	108	216	
				84
4	108	192	300	
				108
5	108	300	408	
				132
6	108	432	540	
				156
7	108	588	696	
				180
8	108	768	876	
				204
9	108	972	1,080	
				228
10	108	1,200	1,308	

Two Key Concepts: Marginal Cost and Average Cost

Marginal Cost

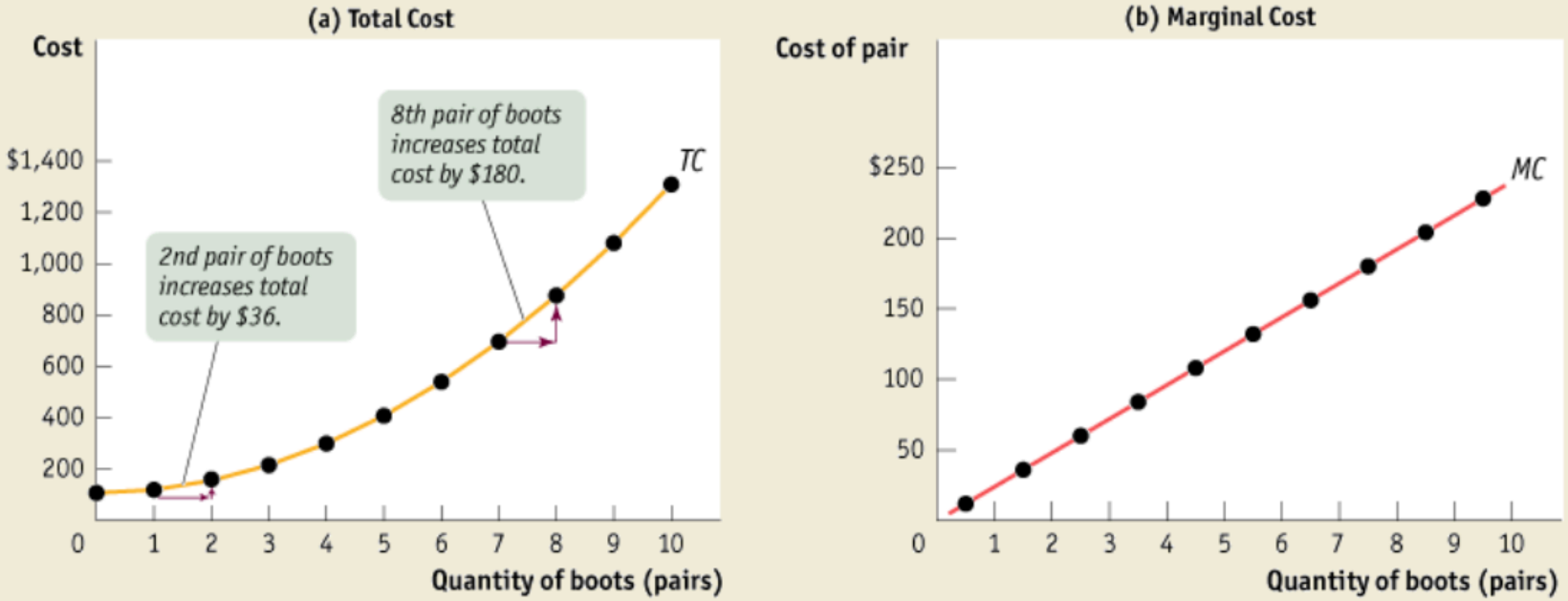
There is **increasing marginal cost** from an activity when each additional unit of the activity costs more than the previous unit.

The **marginal cost curve** shows how the cost of undertaking one more unit of an activity depends on the quantity of that activity that has already been done.

Two Key Concepts: Marginal Cost and Average Cost

Marginal Cost

Figure 7-6 Total Cost and Marginal Cost Curves for Ben's Boots



Panel (a) shows the total cost curve from Table 7-1. Like the total cost curve in Figure 7-4, it slopes upward and gets steeper as we move up it to the right. Panel (b) shows the marginal cost curve. It also slopes upward, reflecting diminishing returns to the variable input.

Two Key Concepts: Marginal Cost and Average Cost

Average Cost

Average total cost, often referred to simply as average cost, is total cost divided by quantity of output produced.

(7-4) $$ATC = \frac{\text{Total cost}}{\text{Quantity of output}} = TC/Q$$

Two Key Concepts:
Marginal Cost and Average Cost

Average Cost

TABLE 7-2

Average Costs for Ben's Boots

Quantity of boots *Q* (pairs)	Fixed cost *FC*	Variable cost *VC*	Total cost *TC*	Average total cost of pair *ATC = TC/Q*	Average fixed cost of pair *AFC = FC/Q*	Average variable cost of pair *AVC = VC/Q*
1	108	12	120	$120.00	$108.00	$12.00
2	108	48	156	78.00	54.00	24.00
3	108	108	216	72.00	36.00	36.00
4	108	192	300	75.00	27.00	48.00
5	108	300	408	81.60	21.60	60.00
6	108	432	540	90.00	18.00	72.00
7	108	588	696	99.43	15.43	84.00
8	108	768	876	109.50	13.50	96.00
9	108	972	1,080	120.00	12.00	108.00
10	108	1,200	1,308	130.80	10.80	120.00

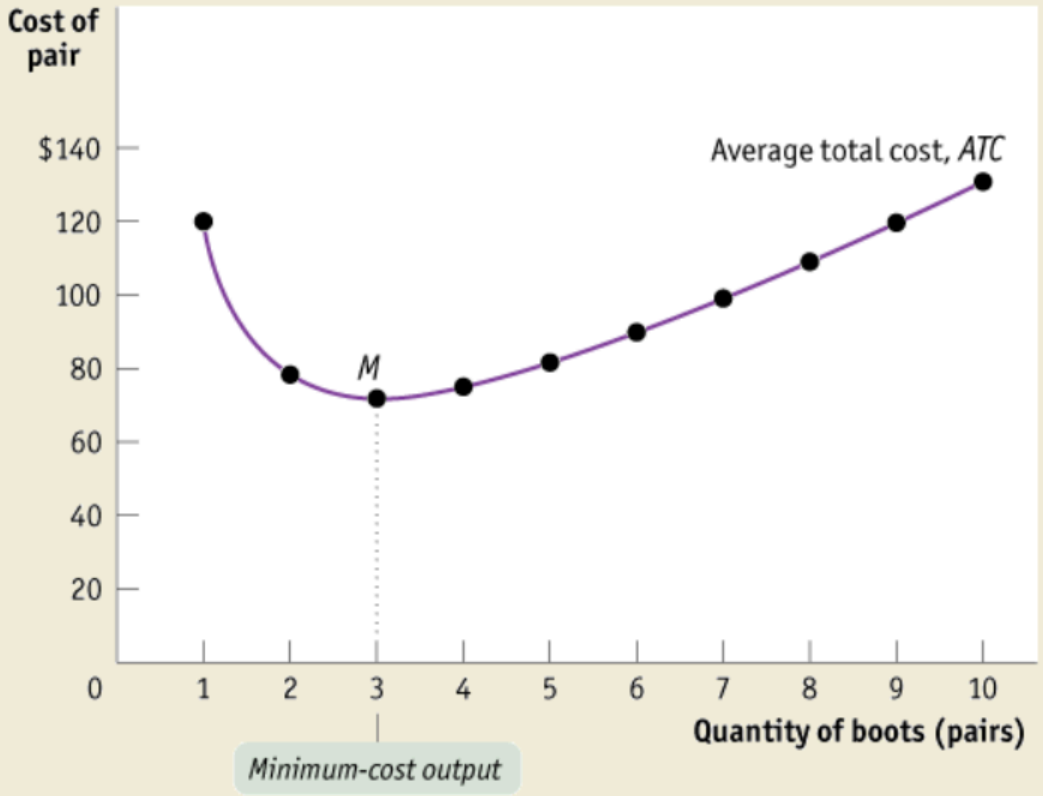
Two Key Concepts: Marginal Cost and Average Cost

Average Cost

Figure 7-7

Average Total Cost Curve for Ben's Boots

The average total cost curve at Ben's Boots is U-shaped. At low levels of output, average total cost falls because the "spreading effect" of falling average fixed cost dominates the "diminishing returns effect" of rising average variable cost. At higher levels of output, the opposite is true and average total cost rises. At point *M*, corresponding to an output of three pairs of boots per day, average total cost is at its minimum level, the minimum average total cost.



Two Key Concepts: Marginal Cost and Average Cost

Average Cost

A **U-shaped average total cost curve** falls at low levels of output, then rises at higher levels.

Average fixed cost is the fixed cost per unit of output.

Average variable cost is the variable cost per unit of output.

(7-5) $$AFC = \frac{\text{Fixed cost}}{\text{Quantity of output}} = FC/Q$$

$$AVC = \frac{\text{Variable cost}}{\text{Quantity of output}} = VC/Q$$

Two Key Concepts: Marginal Cost and Average Cost

Average Cost

Increasing output has two opposing effects on average total cost—the "spreading effect" and the "diminishing returns effect":

- The spreading effect: the larger the output, the more production that can "share" the fixed cost, and therefore the lower the average fixed cost.
- The diminishing returns effect: the more output produced, the more variable input it requires to produce additional units, and therefore the higher the average variable cost.

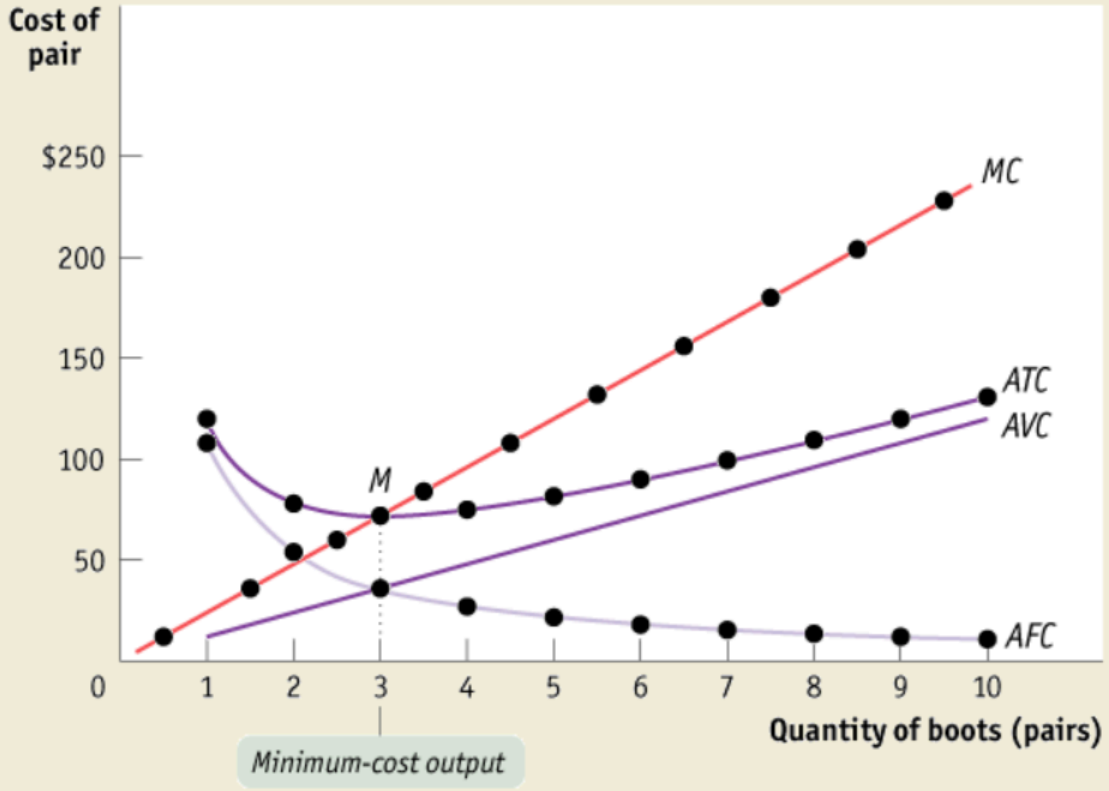
Two Key Concepts: Marginal Cost and Average Cost

Average Cost

Figure 7-8

Marginal Cost and Average Cost Curves for Ben's Boots

Here we have the family of cost curves for Ben's Boots: the marginal cost curve (*MC*), the average total cost curve (*ATC*), the average variable cost curve (*AVC*), and the average fixed cost curve (*AFC*). Note that the average total cost curve is U-shaped and the marginal cost curve crosses the average total cost curve at the bottom of the U, point *M*, corresponding to the minimum average total cost from Table 7-2 and Figure 7-7.



Two Key Concepts:
Marginal Cost and Average Cost

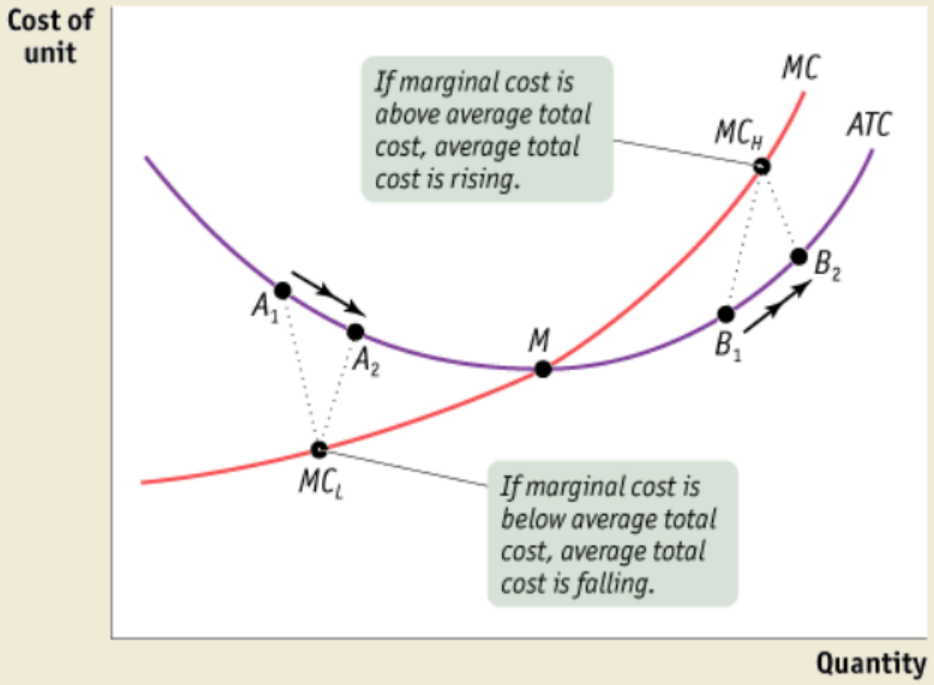
Minimum Average Total Cost

The **minimum-cost output** is the quantity of output at which average total cost is lowest—the bottom of the U-shaped average total cost curve.

Figure 7-9

The Relationship Between the Average Total Cost and the Marginal Cost Curves

To see why the marginal cost curve (*MC*) must cut through the average total cost curve at the minimum average total cost (point *M*), corresponding to the minimum-cost output, we look at what happens if marginal cost is different from average total cost. If marginal cost is less than average total cost, an increase in output must reduce average total cost, as in the movement from A_1 to A_2. If marginal cost is greater than average total cost, an increase in output must increase average total cost, as in the movement from B_1 to B_2.



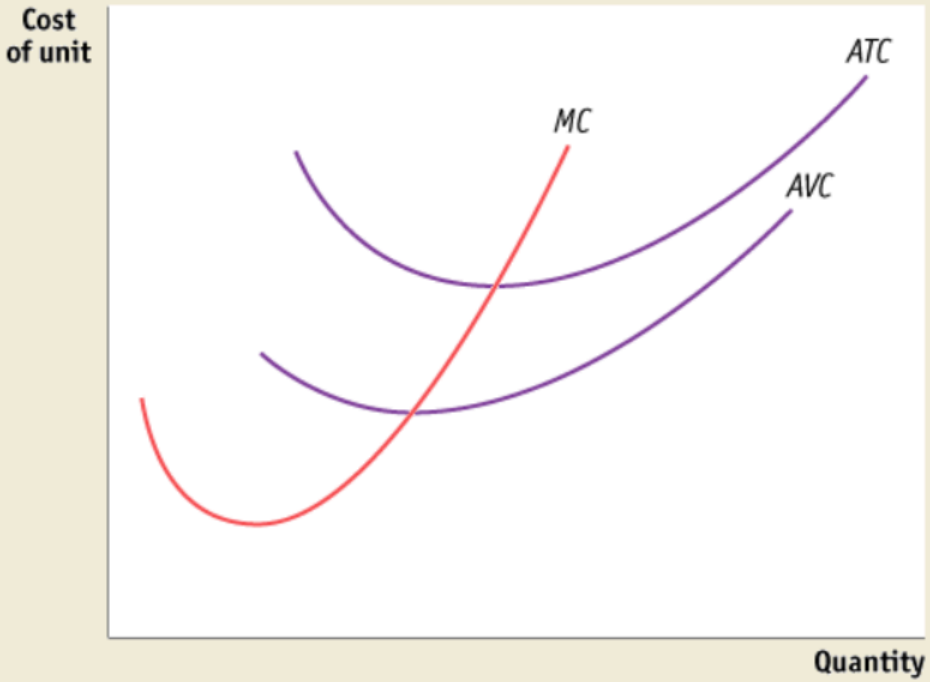
Two Key Concepts:
Marginal Cost and Average Cost

Does the Marginal Cost Curve Always Slope Upward?

Figure 7-10

More Realistic Cost Curves

In practice, the marginal cost curve often begins with a section that slopes downward. As output rises from a low level, a firm is capable of engaging in specialization and division of labor, which leads to increasing returns. At higher levels of output, however, diminishing returns lead to upward-sloping marginal cost. When marginal cost has a downward-sloping section, average variable cost is U-shaped. However, the basic results—U-shaped average total cost, and marginal cost that cuts through the minimum average total cost—remain the same.



Relations among TC, MC, ATC, AVC

TC: Total Cost = TC(Q)
MC: Marginal Cost = $\frac{dTC(Q)}{dQ}$
ATC: Total Average Cost = TC(Q) / Q
AVC: Average Variable Cost = VC(Q) / Q

(1) When MC =ATC, the ATC becomes minimum.

<Proof> For ATC to be minimum, $\frac{dATC(Q)}{dQ} = 0.$

$$\frac{dATC(Q)}{dQ} = \frac{d(\frac{TC(Q)}{Q})}{dQ} = \frac{\frac{dTC(Q)}{dQ} * Q - TC(Q)}{Q^2} = \frac{MC(Q) * Q - TC(Q)}{Q^2}$$

$$= \frac{1}{Q}(MC - \frac{TC(Q)}{Q}) = \frac{1}{Q}(MC - ATC)$$

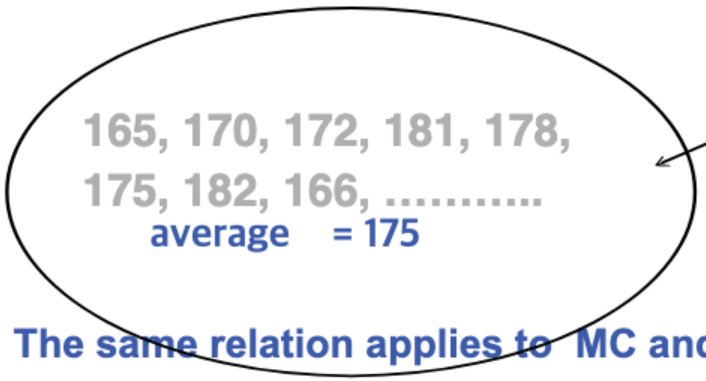
$$So, \quad \frac{dATC(Q)}{dQ} \begin{array}{ll} > 0, & \text{if } MC > ATC, \\ = 0, & \text{if } MC = ATC, \\ < 0, & \text{if } MC < ATC \end{array}$$

Continued - Relations among TC, MC, ATC, AVC

(2) When MC = AVC, the AVC becomes minimum. You can prove as (1).

(3) When MC is less than ATC, the ATC is decreasing.
When MC is greater than ATC, the ATC is increasing.

<Example> From the proof of (1),



For the average to be bigger, the new comer has to be greater than the average

The same relation applies to MC and AVC

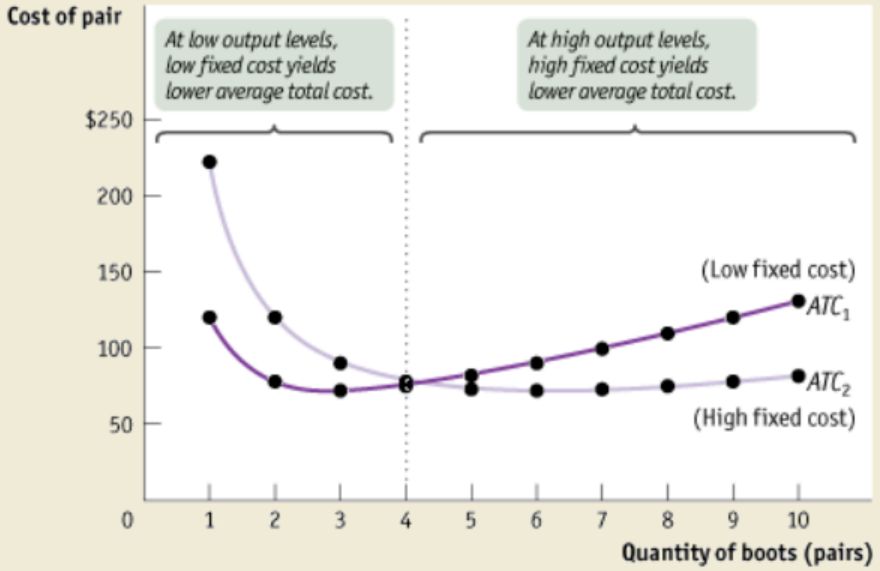
(4) When MC is less than AVC, the AVC is decreasing.
When MC is greater than AVC, the AVC is increasing.

Short-Run versus Long-Run Costs

Figure 7-11

Choosing the Level of Fixed Cost for Ben's Boots

There is a trade-off between higher fixed cost and lower variable cost for any given output level, and vice versa. ATC_1 is the average total cost curve corresponding to a fixed cost of $108; it leads to high variable cost. ATC_2 is the average total cost curve corresponding to a higher fixed cost of $216 but lower variable cost. At low output levels, fewer than 4 pairs of boots per day, ATC_1 lies below ATC_2: average total cost is lower with only $108 in fixed cost. But as output goes up, average total cost is lower with the higher amount of fixed cost, $216: at more than 4 pairs of boots per day, ATC_2 lies below ATC_1.



	Low fixed cost (FC = $108)			High fixed cost (FC = $216)		
Quantity of boots (pairs)	High variable cost	Total cost	Average total cost of pair ATC_1	Low variable cost	Total cost	Average total cost of pair ATC_2
1	$12	$120	$120.00	$6	$222	$222.00
2	48	156	78.00	24	240	120.00
3	108	216	72.00	54	270	90.00
4	192	300	75.00	96	312	78.00
5	300	408	81.60	150	366	73.20
6	432	540	90.00	216	432	72.00
7	588	696	99.40	294	510	72.90
8	768	876	109.50	384	600	75.00
9	972	1,080	120.00	486	702	78.00
10	1,200	1,308	130.80	600	816	81.60

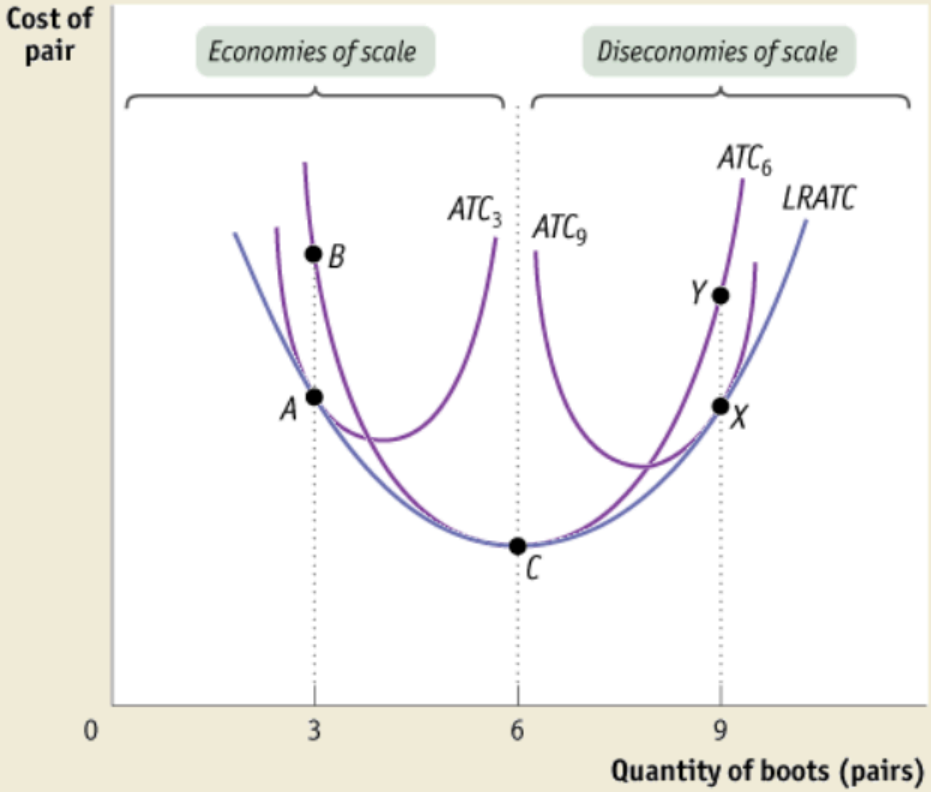
Short-Run versus Long-Run Costs

The **long-run average total cost curve** shows the relationship between output and average total cost when fixed cost has been chosen to minimize average total cost for each level of output.

Figure 7-12

Short-Run and Long-Run Average Total Cost Curves

If Ben's Boots has chosen the level of fixed cost that minimizes short-run average total cost at an output of six pairs of boots per day and actually ends up producing six pairs per day, it will be on *LRATC* at point *C*. But if it produces more or less, in the short run it will be on the short-run average total cost curve ATC_6 and not on *LRATC*. So if it produces only three pairs per day, its average total cost is shown by point *B*, not point *A*. If it produces nine pairs per day, its average total cost is shown by point *Y*, not point *X*. There are economies of scale when long-run average total cost declines as output increases, and there are diseconomies of scale when long-run average total cost increases as output increases.



Short-Run versus Long-Run Costs

Economies and Diseconomies of Scale

There are **economies of scale** when long-run average total cost declines as output increases.

There are **diseconomies of scale** when long-run average total cost increases as output increases.

There are **constant returns to scale** when long-run average total cost is constant as output increases.

Short-Run versus Long-Run Costs

Summing Up Costs: The Short and Long of It

TABLE 7-3

Concepts and Measures of Cost

	Measurement	Definition	Mathematical term
Short run	Fixed cost	Cost that does not depend on the quantity of output produced	*FC*
	Average fixed cost	Fixed cost per unit of output	*AFC = FC/Q*
Short run and long run	Variable cost	Cost that depends on the quantity of output produced	*VC*
	Average variable cost	Variable cost per unit of output	*AVC = VC/Q*
	Total cost	The sum of fixed cost (short-run) and variable cost	*TC = FC* (short-run) *+ VC*
	Average total cost (average cost)	Total cost per unit of output	*ATC = TC/Q*
	Marginal cost	The change in total cost generated by producing one more unit of output	*MC = ΔTC/ΔQ*
Long run	Long-run average total cost	Average cost when fixed cost has been chosen to minimize total cost for each level of output	*LRATC*

KEY TERMS

Production function
Fixed input
Variable input
Long run
Short run
Total product curve
Marginal product
Diminishing returns to an input
Implicit cost
Explicit cost
Fixed cost
Variable cost
Total cost
Total cost curve
Marginal cost
Increasing marginal cost
Marginal cost curve
Average total cost
Average cost
U-shaped average total cost curve
Average fixed cost
Average variable cost
Minimum-cost output
Long-run average total cost curve
Economies of scale
Diseconomies of scale
Constant returns to scale